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degli Studi Sociali Guido Carli

# Homework Exercises on Binary Numbers and Boolean Logic

## Introduction to Computer Programming

**Management and Artificial Intelligence**

# Binary Number Exercises

**Challenging homework problems!**

# Decimal 387 to Binary

Convert the decimal number **387** to binary:

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$387 \div 2 = 193$ (rem 1)

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$387 \div 2 = 193$ (rem 1)

$193 \div 2 = 96$ (rem 1)

$96 \div 2 = 48$ (rem 0)

# Decimal 387 to Binary

Convert the decimal number **387** to binary:

$387 \div 2 = 193$ (rem 1)

$193 \div 2 = 96$ (rem 1)

$96 \div 2 = 48$ (rem 0)

$48 \div 2 = 24$ (rem 0)

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Convert the decimal number **387** to binary:

$387 \div 2 = 193$ (rem 1)

$193 \div 2 = 96$ (rem 1)

$96 \div 2 = 48$ (rem 0)

$48 \div 2 = 24$ (rem 0)

$24 \div 2 = 12$ (rem 0)

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$387 \div 2 = 193$ (rem 1)

$193 \div 2 = 96$ (rem 1)

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$48 \div 2 = 24$ (rem 0)

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

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$48 \div 2 = 24$ (rem 0)

$24 \div 2 = 12$ (rem 0)

$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

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$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

$3 \div 2 = 1$ (rem 1)

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$12 \div 2 = 6$ (rem 0)

$6 \div 2 = 3$ (rem 0)

$3 \div 2 = 1$ (rem 1)

$1 \div 2 = 0$ (rem 1)

Result: $(110000011)_2$

# Binary 110110101011 to Decimal

Convert the binary number **110110101011** to decimal:

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Convert the binary number **110110101011** to decimal:

$$(110110101011)_2 = 1 \cdot 2^{11} + 1 \cdot 2^{10} + 0 \cdot 2^9 + 1 \cdot 2^8 + 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 -$$

$$= 2048 + 1024 + 0 + 256 + 128 + 0 + 32 + 0 + 8 + 0 + 2 + 1 = (3499)_{10}$$

# Decimal 2048 to Hexadecimal

Convert the decimal number **2048** to hexadecimal:

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Convert the decimal number **2048** to hexadecimal:

$2048 \div 16 = 128$ remainder $0$

$128 \div 16 = 8$ remainder $0$

$8 \div 16 = 0$ remainder $8$

Reading the remainders from bottom to top: $800$

Therefore, $2048_{10} = 800_{16}$

# Hex BEEF to Binary

Convert the hexadecimal number **BEEF** to binary:

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Convert the hexadecimal number **BEEF** to binary:

Convert each hexadecimal digit to its 4-bit binary equivalent:

- $B_{16} = 11_{10} = 1011_2$
- $E_{16} = 14_{10} = 1110_2$
- $E_{16} = 14_{10} = 1110_2$
- $F_{16} = 15_{10} = 1111_2$

Therefore: $BEEF_{16} = 1011111011101111_2$

# Octal 1777 to Decimal

Convert the octal number **1777** to decimal:

# Octal 1777 to Decimal

Convert the octal number **1777** to decimal:

$$(1777)_8 = 1 \cdot 8^3 + 7 \cdot 8^2 + 7 \cdot 8^1 + 7 \cdot 8^0$$

$$= 1 \cdot 512 + 7 \cdot 64 + 7 \cdot 8 + 7 \cdot 1$$

$$= 512 + 448 + 56 + 7 = (1023)_{10}$$

# Boolean Logic Exercises

Formula $(A \wedge \neg B \wedge C) \vee (\neg A \wedge B \wedge \neg C) \vee (A \wedge B \wedge C)$

Formula $(A \land \neg B \land C) \lor (\neg A \land B \land \neg C) \lor (A \land B \land C)$

| A | B | C | !A | !B | !C | A∧!B∧C | !A∧B∧!C | A∧B∧C | (A∧!B∧C)∨(!A∧B∧!C)∨(A∧B∧C) |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |

Formula $(\neg A \wedge \neg B \wedge \neg C) \vee (A \wedge B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

Formula $(\neg A \wedge \neg B \wedge \neg C) \vee (A \wedge B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

| A | B | C | !A | !B | !C | !A∧!B∧!C | A∧B∧C | A∧!B∧!C | (!A∧!B∧!C)∨(A∧B∧C)∨(A∧!B∧!C |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |

Formula $(A \wedge B) \vee (\neg A \wedge \neg B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

Formula $(A \wedge B) \vee (\neg A \wedge \neg B \wedge C) \vee (A \wedge \neg B \wedge \neg C)$

| A | B | C | !A | !B | !C | A∧B | !A∧!B∧C | A∧!B∧!C | (A∧B)∨(!A∧!B∧C)∨(A∧!B∧!C) |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |

Formula $(A \land B \land C) \lor (\neg A \land \neg B \land \neg C)$ is equivalent to what?

Formula $(A \wedge B \wedge C) \vee (\neg A \wedge \neg B \wedge \neg C)$ is equivalent to what?

| A | B | C | A∧B∧C | !A | !B | !C | !A∧!B∧!C | (A∧B∧C)∨(!A∧!B∧!C) |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |

# Formula $(A \wedge B \wedge C) \vee (\neg A \wedge \neg B \wedge \neg C)$ is equivalent to what?

| A | B | C | A∧B∧C | !A | !B | !C | !A∧!B∧!C | (A∧B∧C)∨(!A∧!B∧!C) |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |

**This formula is true when all variables have the same value (all true or all false)**

Formula $(A \vee B) \wedge \neg(A \wedge B)$ is a tautology?

Formula $(A \vee B) \wedge \neg(A \wedge B)$ is a tautology?

| A | B | A∨B | A∧B | !(A∧B) | (A∨B)∧!(A∧B) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 |

Formula $(A \vee B) \wedge \neg(A \wedge B)$ is a tautology?

| A | B | A∨B | A∧B | !(A∧B) | (A∨B)∧!(A∧B) |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 |

**No, it is not a tautology! This is actually the XOR operation.**

Formula $\neg A \land \neg B \land A$ is a contradiction?

Formula $\neg A \wedge \neg B \wedge A$ is a contradiction?

| A | B | !A | !B | !A∧!B∧A |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 |

Formula $\neg A \wedge \neg B \wedge A$ is a contradiction?

| A | B | !A | !B | !A∧!B∧A |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 |

**Yes, it is a contradiction! It requires A to be both true and false simultaneously.**